

AJAY GANESH

Software Engineer

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Education

University of Michigan

Ann Arbor, MI

Bachelors of Science in Computer Science, 3.73 GPA

Graduated May 2024

- **Course Highlights:** Algorithms and Data Structures, Web Systems, Operating Systems, Distributed Systems, Computer Security, Compiler Construction, Discrete Mathematics, Machine Learning and Practical Data Science

Technical Skills

Languages: Python, C++, Rust, Go, JavaScript/TypeScript, HTML/CSS, SQL

Technologies: React, MongoDB, PostgreSQL, Node.js, Git, Docker, Linux, AWS

Experience

Meta

Menlo Park, CA

Software Engineer

June 2024 - Present

- Currently working on the Applied AI Agent Data Optimization team to develop data and evaluation infrastructure supporting the training of Meta's frontier generative AI/LLM efforts
- Previously worked on the Ads Privacy Infra team, developing infrastructure in C++, Python, and Hack/PHP to support Meta's ads systems and their compliance with privacy regulations, safeguarding the data of billions of users and advertisers worldwide across Facebook, Instagram, and WhatsApp

Deepgram

Ann Arbor, MI

Software Engineering Intern

May 2023 - Aug. 2023

- Developed and deployed a Node.js backend service enabling speech recognition machine learning model usage on YouTube videos, used in a high-value client demo with potential revenue exceeding \$100k
- Optimized code for a Rust web API handling authorization and billing for Deepgram's Natural Language Processing models, resulting in a 5 ms average decrease in response time and greater standardization of error messages
- Increased account signup conversions by 7% by implementing an IP caching system in Rust and deploying it to on-premises data centers, giving new users increased exposure to Deepgram's AI models

Riot Games

Ann Arbor, MI

Student Security Engineer - UM MDP Program

Jan. 2023 - Present

- Developed an extensible monitoring system using Python and Go to provide security engineers with real-time notifications for vulnerabilities in internal networks, contributing to increased operational efficiency
- Harnessed Amazon Web Services (Lambda, S3, CloudFormation) to create a deployment pipeline, allowing engineers to automatically deploy internal environment monitoring scripts in minutes
- Researched workflows with senior Information Security professionals to develop a continuous compliance dashboard in React, allowing executives to easily visualize the status of the organization's attack surface

CaringWire

Columbus, OH

Software Engineering Intern

Nov. 2021 - May 2022

- Collaborated with senior developers and interns in an Agile environment to make updates and improvements to a web platform using Scrum ideologies, improving user experience for hundreds of caretakers and families
- Implemented features such as an auto-logout system, a database search autocomplete feature, and a cloud upload system for medical data, achieving HIPAA compliance and increasing usability for elderly consumers
- Improved a search feature by implementing a MongoDB data aggregation pipeline using TypeScript, leading to a 75% increase in quantity of accurate results delivered to users

Personal Projects

Distributed Key-Value Database

Oct. 2023

- Tech Stack: Go, RPCs, AWS
- Leveraged the Paxos consensus algorithm to create a sharded key-value store in Golang, ensuring high consistency, fault-tolerance, and load balancing; further optimized by reducing network communication and custom garbage collection

C++ Thread Library

Jan. 2023

- Tech Stack: C++, Make, POSIX Threads
- Architected and Implemented a thread library from scratch in C++ to allow for greater concurrency, including synchronization through primitives such as locks and condition variables